

Chapter 7: Special Forms of Matrices

Example: Consider $M = \begin{bmatrix} 1 & 3 & 0 \\ -2 & 4 & 10 \\ 0 & 1 & -1 \end{bmatrix}$.

Compute $3M$.

$$3M = \begin{bmatrix} 3 & 9 & 0 \\ -6 & 12 & 30 \\ 0 & 3 & -3 \end{bmatrix}$$

Is there a 3×3 matrix C where $CM = 3M$?

$$C = 3I_3$$

Definition: An $n \times n$ matrix A is a **scalar matrix** if $A = cI_n$ for some scalar c (remember: **scalar** means the same thing as **real number**).

Definition: An $n \times n$ matrix $A = [a_{ij}]$ is called **upper triangular** if $a_{ij} = 0$ for $i > j$, and A is called **lower triangular** if $a_{ij} = 0$ for $i < j$. A diagonal matrix is both upper and lower triangular.

Example: $A = \begin{bmatrix} 0 & 0 & 0 \\ 4 & 2 & 0 \\ -1 & 1 & 1 \end{bmatrix}$ is lower triangular, while $B = \begin{bmatrix} 3 & 2 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}$ is upper triangular.

Example: Consider $D = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$.

What is D^4 ?

$$D^4 = \begin{bmatrix} a^4 & 0 \\ 0 & b^4 \end{bmatrix}$$

If D is invertible, what is D^{-1} ?

$$D^{-1} = \begin{bmatrix} a^{-1} & 0 \\ 0 & b^{-1} \end{bmatrix}$$

Definition: A matrix A with real entries is called **symmetric** if $A^T = A$. A matrix A is called **skew symmetric** if $A^T = -A$.

Example: Are the following matrices symmetric/skew symmetric/neither?

$$A = \begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & 5 \\ 3 & -5 & 0 \end{bmatrix} \quad a_{ij} = -a_{ji} \text{ for } \forall i, j. \Rightarrow \text{skew-sym.}$$

$$B = \begin{bmatrix} 3 & 5 & 2 \\ 5 & 1 & 4 \\ 2 & 4 & -1 \end{bmatrix} \quad b_{ij} = b_{ji} \text{ for } \forall i, j \Rightarrow \text{symmetric matrix}$$

$$C = \begin{bmatrix} 1 & 2 & -3 \\ -2 & 0 & 5 \\ 3 & 5 & 0 \end{bmatrix} \quad \text{neither } C_{ij} = C_{ji} \text{ or } C_{ij} = -C_{ji}. \\ \Rightarrow \text{neither sym. or skew-sym.}$$

$$D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad d_{ij} = \pm d_{ji} \text{ for } \forall i, j. \Rightarrow \text{both sym.}$$

Note: If a matrix A is symmetric and skew symmetric, then A must be a **zero** matrix! & skew-sym.

Algebra of Special Forms We end by summarizing some of the algebra of these special forms.

Theorem:

1. The result of adding two *diagonal* matrices, scalar multiplying a *diagonal* matrix, multiplying two *diagonal* matrices, taking the inverse of a *diagonal* matrix (if invertible), or taking a power (with positive or negative exponent, if invertible) of a *diagonal* matrix is always a *diagonal* matrix.
2. Statement 1 remains true when (a) all occurrences of *diagonal* are replaced by *scalar*, (b) all occurrences of *diagonal* are replaced by *upper triangular*, or (c) all occurrences of *diagonal* are replaced by *lower triangular*.
3. Statement 1 remains true when all occurrences of *diagonal* are replaced by *symmetric*, except that the product of two symmetric matrices may not be symmetric.

Question: Which of the properties are true for skew-symmetric matrices?

→ Proof of statement 2:

Assume A, B are upper triangular (lower triangular case is similar).

then $a_{ij} = b_{ij} = 0$ for $\forall i > j$.

$$(i) (A+B)_{ij} = a_{ij} + b_{ij} = 0 \text{ for } \forall i > j.$$

$\Rightarrow A+B$ is upper tri.

$$(ii) (\lambda A)_{ij} = \lambda a_{ij} = 0 \text{ for } \forall i > j.$$

$$(iii) (A \cdot B)_{ij} = \underbrace{a_{i1}}_{=0} \cdot \underbrace{b_{1j}}_{=0} + \underbrace{a_{i2}}_{=0} \cdot \underbrace{b_{2j}}_{=0} + \dots + \underbrace{a_{i,i-1}}_{=0} \cdot \underbrace{b_{i-1,j}}_{=0} \\ + \underbrace{a_{ii}}_{=0} \cdot \underbrace{b_{ij}}_{=0} + \underbrace{a_{i,i+1}}_{=0} \cdot \underbrace{b_{i+1,j}}_{=0} + \dots + \underbrace{a_{in}}_{=0} \cdot \underbrace{b_{nj}}_{=0}$$

(iv). A^{-1} and (v). A^k will be left as exercises.

Chapter 8: Determinants

Go to OH if have questions.

In Chapters 8-10, we are working with square $n \times n$ matrices.

Goal of these chapters: Find a number (called a **determinant**) that we can calculate from a matrix that tells us whether or not it's invertible.

Let's start with 2×2 : $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

We saw a result in a previous chapter that

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

When can this go wrong?

Definition: The **determinant** of $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is

$$\det(A) = ad - bc.$$

Example: Find the determinant of $A = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 2 \\ 6 & 4 \end{bmatrix}$

$$\det A = 2 \quad , \quad \det B = 0.$$

We can use the 2×2 determinant to build the 3×3 determinant. It will *not* be clear right now why this is the correct analogous function, but we'll explore that more later.

Definition: Let A be a 3×3 matrix. The i, j^{th} **minor** of A , denoted $\text{minor}(A)_{ij}$, is the determinant of the 2×2 matrix which results from deleting the i^{th} row and j^{th} column of A .

In general, if A is $n \times n$, then the i, j^{th} minor of A is the determinant of the $(n-1) \times (n-1)$ matrix which results from deleting the i^{th} row and j^{th} column of A .

Example: Find $\text{minor}(A)_{11}$ and $\text{minor}(A)_{23}$ for

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 3 & 2 & 1 \end{bmatrix}$$

$$\text{minor}(A)_{11} = \det \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}$$

$$\text{minor}(A)_{23} = \det \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$$

Definition: Let A be a $n \times n$ matrix. The i, j^{th} **cofactor** of A , denoted $\text{cof}(A)_{ij}$, is the defined to be

$$\text{cof}(A)_{ij} = (-1)^{i+j} \text{minor}(A)_{ij}$$

Example: Find $\text{cof}(A)_{11}$ and $\text{cof}(A)_{23}$ for

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 3 & 2 & 1 \end{bmatrix}$$

$$\text{cof}(A)_{11} = (-1)^2 \text{minor}(A)_{11} = \det \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}$$

$$\text{cof}(A)_{23} = (-1)^5 \text{minor}(A)_{23} = - \det \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}.$$

Definition: Let A be a 3×3 matrix. Then the **determinant** of A , $\det(A)$, is calculated by picking a row (or column), taking the product of each entry of that row (column) with its cofactor, then adding these products together. In other words, if we pick row i , this process is called **expanding along the i th row** and is given as

$$\det(A) = a_{i1} \text{cof}(A)_{i1} + a_{i2} \text{cof}(A)_{i2} + a_{i3} \text{cof}(A)_{i3}$$

This process is called **calculating the determinant by cofactor expansion**.

Example: Find the determinant of

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 3 & 2 & 1 \end{bmatrix}$$

by expanding along row 1. Also try expanding along column 2.

Row 1 expansion:

$$\det A = 1 \cdot \det \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix} - 2 \det \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix} + 3 \cdot \det \begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix}$$

Col. 2 expansion:

$$\det A = -2 \det \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix} + 3 \cdot \det \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix} - 2 \det \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix}$$

To track the sign $(-1)^{i+j}$, we can view the matrix as a checkerboard pattern:

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

Note: Since the row/column doesn't matter, you should choose the "easiest" row/column:

Example:

$$\det \left(\begin{bmatrix} 3 & -5 & 1 \\ 0 & 1 & 0 \\ -2 & -4 & 7 \end{bmatrix} \right) = 1 \cdot \det \begin{bmatrix} 3 & 1 \\ -2 & 7 \end{bmatrix} = 23.$$

$$\det \left(\begin{bmatrix} 0 & \pi & 200 \\ 0 & 0.03 & 4 \\ 0 & 19 & -3584 \end{bmatrix} \right) = 0.$$

We can compute the determinant of a general $n \times n$ matrix by recursively repeating this process.

Example: $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 4 & 2 & 3 \\ 1 & 3 & 4 & 5 \\ 3 & 4 & 3 & 2 \end{bmatrix}$, if we expand along third column, we get

$$\det(A) = 3(-1)^{1+3} \det \begin{pmatrix} \begin{bmatrix} 5 & 4 & 3 \\ 1 & 3 & 5 \\ 3 & 4 & 2 \end{bmatrix} \end{pmatrix} + 2(-1)^{2+3} \det \begin{pmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 1 & 3 & 5 \\ 3 & 4 & 2 \end{bmatrix} \end{pmatrix} + 4(-1)^{3+3} \det \begin{pmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 5 & 4 & 3 \\ 3 & 4 & 2 \end{bmatrix} \end{pmatrix} + 3(-1)^{4+3} \det \begin{pmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 5 & 4 & 3 \\ 1 & 3 & 5 \end{bmatrix} \end{pmatrix} = -12.$$

Definition: Let A be a $n \times n$ matrix ($n \geq 3$). Then the **determinant** of A , $\det(A)$, is

$$\det(A) = \sum_{j=1}^n a_{ij} \operatorname{cof}(A)_{ij} = \sum_{i=1}^n a_{ij} \operatorname{cof}(A)_{ij}$$

for some choice of row i or column j .

Example: Compute $\det \begin{bmatrix} 1 & -1 & 0 & 3 \\ 4 & 4 & 2 & 2 \\ 0 & 8 & 0 & -1 \\ 0 & 9 & 0 & 1 \end{bmatrix} = -2 \times \det \begin{bmatrix} 1 & -1 & 3 \\ 0 & 8 & -1 \\ 0 & 9 & 1 \end{bmatrix}$

$$= -2 \times 1 \times \det \begin{bmatrix} 8 & -1 \\ 9 & 1 \end{bmatrix} = -2 \times (8 + 9) = -34$$

Example: Compute $\det \begin{bmatrix} 3 & 2 & 3 & 4 \\ 0 & 4 & 2 & 3 \\ 0 & 0 & 2 & 5 \\ 0 & 0 & 0 & -1 \end{bmatrix}$

$$= 3 \cdot \det \begin{bmatrix} 4 & 2 & 3 \\ 0 & 2 & 5 \\ 0 & 0 & -1 \end{bmatrix} = 3 \times 4 \cdot \det \begin{bmatrix} 2 & 5 \\ 0 & -1 \end{bmatrix} = 3 \times 4 \times 2 \times (-1) = -24$$

Theorem: If $A = [a_{ij}]$ is a upper/lower triangular matrix, then $\det(A) = a_{11}a_{22} \cdots a_{nn}$, that is, the determinant of a triangular matrix is the product of the elements on the main diagonal.

Chapter 9: Determinants versus Row Operations

Example: The matrix $A = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 2 & -2 \\ -4 & 3 & 1 \end{bmatrix}$ reduces to $\begin{bmatrix} 1 & 1 & 2 \\ 0 & -1 & 4 \\ 0 & 0 & 25 \end{bmatrix}$ via elementary row operations. Fill in each row operation.

$$\begin{bmatrix} 3 & 2 & 1 \\ 2 & 2 & -2 \\ -4 & 3 & 1 \end{bmatrix} \xrightarrow[\times(-1)]{r_1 \leftrightarrow r_2} \begin{bmatrix} 2 & 2 & -2 \\ 3 & 2 & 1 \\ -4 & 3 & 1 \end{bmatrix} \xrightarrow[\times(\frac{1}{2})]{r_1 \rightarrow \frac{1}{2}r_1} \begin{bmatrix} 1 & 1 & -1 \\ 3 & 2 & 1 \\ -4 & 3 & 1 \end{bmatrix}$$

$$\xrightarrow[r_3 \rightarrow r_3 + 4r_1]{r_2 \rightarrow r_2 - 3r_1} \begin{bmatrix} 1 & 1 & -1 \\ 0 & -1 & 4 \\ 0 & 7 & -3 \end{bmatrix} \xrightarrow[\times 1]{r_3 \rightarrow r_3 + 7r_2} \begin{bmatrix} 1 & 1 & -1 \\ 0 & -1 & 4 \\ 0 & 0 & 25 \end{bmatrix}$$

The determinant of $\begin{bmatrix} 1 & 1 & 2 \\ 0 & -1 & 4 \\ 0 & 0 & 25 \end{bmatrix}$ is easy to compute. Can we say something about the determinant of A knowing the row operations that led us to this matrix?

Properties of Determinants

Theorem: If A is a square matrix, then $\det(A) = \det(A^T)$.

Justification chat: (i.e. "proof"):

Heuristic: Since the area (volume) spanned by the col. vectors of A and A^T are the same.

Theorem: If B results from A by interchanging two different rows (or two different columns), then $\det(B) = -\det(A)$.

Theorem: If two rows (columns) of A are equal, then $\det(A) = 0$.

Proof:

Assume i^{th} & j^{th} row of A are equal.

Let B be the matrix obtained by interchanging i^{th} & j^{th} row of A , so $B = A$. By the

previous thm, $\det(\underline{B}) = -\det(A) \Rightarrow \det(A) = 0$.

Theorem: If a row (column) of A consists entirely of zeros, then $\det(A) = 0$.

Proof:

Say the i^{th} row (column) of A is zero,
row (column) expand the i^{th} row (column).

$$\det(A) = \underbrace{a_{i1}}_0 \cdot \text{cof}(A)_{i1} + \underbrace{a_{i2}}_0 \cdot \text{cof}(A)_{i2} + \cdots + \underbrace{a_{in}}_0 \cdot \text{cof}(A)_{in}$$

Theorem: If B is obtained from A by multiplying a row (column) of A by a real number k , then $\det(B) = k \det(A)$.

Proof:

Let $B = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ k a_{i1} & \cdots & k a_{in} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}$, row expand the i^{th} row of B .

$$\det B = k a_{i1} \cdot \text{cof}(B)_{i1} + \cdots + k a_{in} \cdot \text{cof}(B)_{in}$$

Fact: If A is an $n \times n$ matrix, then $\det(kA) = \cancel{k^n \det(A)} = k \det(A)$.

Theorem: If B is obtained by multiplying the i^{th} row of A by k and adding it to the j^{th} row of A ($i \neq j$) then $\det(A) = \det(B)$.

Check for 2×2 :

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

$$\det \begin{pmatrix} a & b \\ ka+c & kb+d \end{pmatrix} = a \cdot (kb+d) - b \cdot (ka+c)$$

$$= akb + ad - bka - bc$$

$$= ad - bc.$$

Row Operations and the Determinant: Here is a summary of how the elementary row operations affect the determinant:

1. Type I: $\det(A_{kr_i \rightarrow r_i}) = k \det(A)$
2. Type II: $\det(A_{r_i \leftrightarrow r_j}) = -\det(A) \quad (i \neq j)$
3. Type III: $\det(A_{kr_i + r_j \rightarrow r_j}) = \det(A) \quad (i \neq j)$

Example: If B is obtained from A by the following operations: $r_1 \leftrightarrow r_2$, $4r_4 \rightarrow r_4$, and $3r_2 + r_3 \rightarrow r_3$, and $\det(A) = 8$, what is the determinant of B ?

$$\det(A) \xrightarrow[r_1 \leftrightarrow r_2]{\times (-1)} \det(A_1) \xrightarrow[r_4 \rightarrow 4r_4]{\times 4} \det(A_2) \xrightarrow[r_3 \rightarrow r_3 + 3r_2]{\times 1} \det(B)$$

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$$\det(B) = 8 \times (-1) \times 4 \times 1 = -32$$

Example: Go back to the example with $A = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 2 & -2 \\ -4 & 3 & 1 \end{bmatrix}$. What is the determinant of A ?

Example: Compute the determinant of the matrix $B = \begin{bmatrix} 1 & 2 & 1 \\ -1 & 3 & 3 \\ 0 & 15 & 12 \end{bmatrix}$

$$\det A \times (-1) \times \left(\frac{1}{2}\right) \times 1 \times 1 = -25$$

$$\Rightarrow \det A = 50$$